

LivehouseVR: Creating an Immersive and Inclusive Social VR Experience for DJs and Audiences towards Gender Equality

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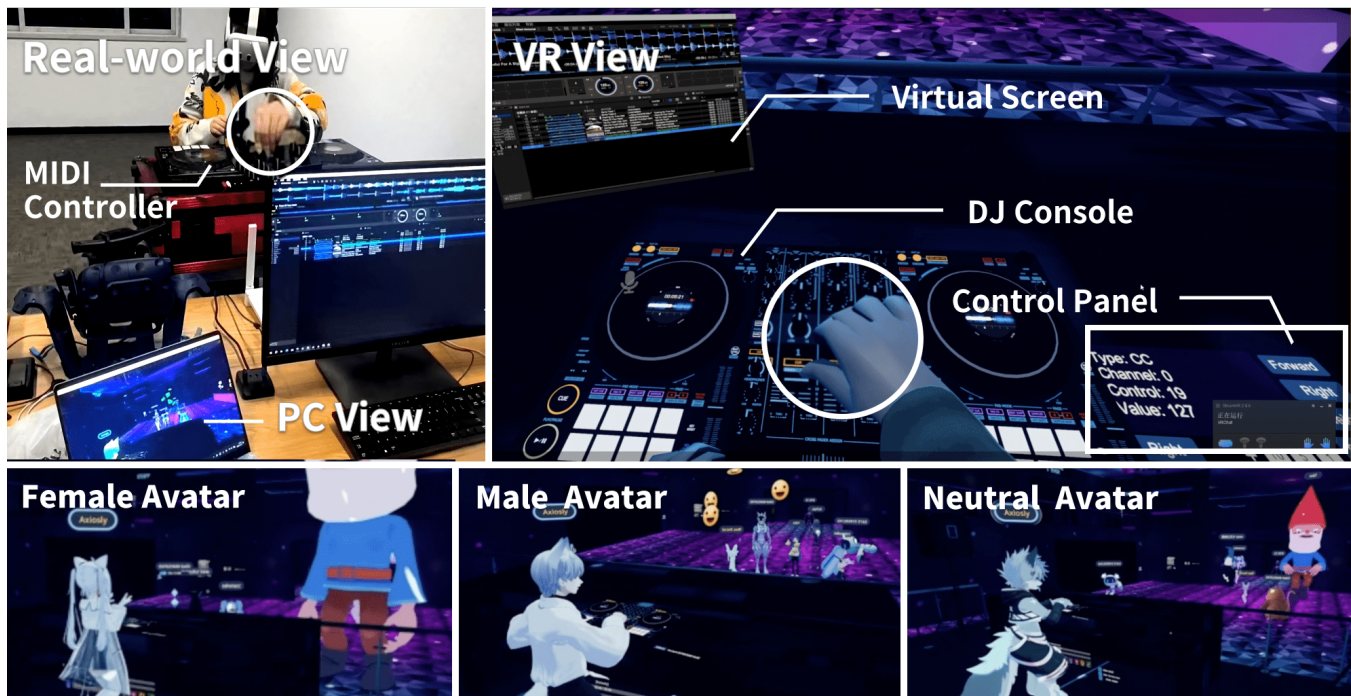


Figure 1: Demonstration of LivehouseVR: real-world environment, user interfaces, and avatar representation.

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Abstract

Gender equality is a fundamental human right, essential for building a peaceful, prosperous, and sustainable community. Despite the progress made over the last decades, old-fashioned sexist attitudes are still common in many professional fields, including DJing. Virtual Reality offers a potential solution for reducing bias by hiding the biological gender identity, thereby shifting the public focus from human features and stereotypes to DJ's quality. In this paper, we design and curate an interactive live house event in VRChat to explore the role of avatar gender in

shaping social interactions and performer identity in DJ performances. Additionally, we integrate a rich array of features to support seamless interaction between the real and virtual worlds in live house events. Preliminary findings from user feedback indicated that our design solution has promising potential to promote gender equality and inclusive interaction. This contributes to a more equitable DJing community aligned with sustainable development goals.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Virtual reality**; **User centered design**.

Keywords

virtual reality, djing, gender equality, live house, sustainable development

ACM Reference Format:

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1 INTRODUCTION

DJing is a highly expressive musical performance that emphasizes interaction with the music, environment and audiences, fostering a vibrant social culture within a dynamic musical scene. The rapid development of affordable midi controllers and visualized DJ mixing software have brought accessible ways for music composition to practitioners. However, implicit bias still exists in DJ community, lacking in diversity and inclusion [10]. Female DJs always face challenges, such as unequal pay in club activities, limited opportunities to be in festival lineups, and old stereotypes of their professional skills. A recent report by DJMag revealed that a mere 12% of DJs are female in the top 100 best world DJs [5], indicating an urgent need for females to break the barriers and gain recognition for their talents on stage.

In this context, virtual reality (VR) is expected to become a promising technical means for mitigating implicit gender biases. A body of prior work has demonstrated its effectiveness in eliciting perceptual illusion of presence in non-physical settings [7, 8, 13]. This involves the ingenious design of external audio and visual cues in a virtual environment to isolate the impact of specific factors, like gender, thereby affecting user behavior and perception [11]. In addition, Yee and Bailenson [21] suggested that user behaviors in virtual world tend to conform to the digital self-representation of their avatars, which is independent of other people's perception in social interactions. On the other hand, the anonymity of avatar representation can help reduce negative stereotypes, prejudice and stigmatization, facilitating more friendly interactions [12].

In terms of creating an immersive DJ experience, it is crucial to focus on how VR technology can be applied to DJing. In recent years, virtual dance [15], drinking [3] and DJ culture [20] in VRChat has been gaining traction. Meanwhile, some emerging

virtual DJ software like Tribe XR has enabled users to mix music on virtual decks and controllers with simple visual instructions. Thus, social VR space could be a viable alternative to traditional live house environments. However, these existing solutions lack the tactile feedback and actual control from physical devices to embody interactions similar to real-world scenarios. This leads to DJs' capability, individuality and creativity being restricted in this way, especially for those familiar with traditional settings and responses. Furthermore, nonverbal communication plays a vital role in the DJ performance. This mainly relies on visible body language or a visual cue [14] to establish an emotional connection with audiences, which requires further consideration for creating a vibrant atmosphere in social VR.

Our research aims to overcome gender bias by creating an immersive and inclusive social VR experience that allows DJs to fully utilize their talents. Through semi-structured expert interviews and an online survey with the public, we identified five key design goals in a user journey. Drawing from these insights, we curate an interactive live house event on a mainstream social VR platform - VRChat, to conduct a user study. Results show that our approach to curating a social VR performance can significantly eliminate gender effects, and facilitate more engaging interactions between DJs and audiences. This research contributes to a deeper understanding of the interplay between technology, gender, and creative interaction. We discuss these efforts moving towards sustainable development goals, and conclude our work.

2 REQUIREMENT GATHERING

2.1 Expert Interview

To understand the feasibility of applying VR technology for DJing and professional insights about gender bias in the DJ industry towards our target groups, we first conducted expert interviews with three practitioners, 2 females and 1 male (age $M=25.00$, $SD=1.63$). It is extremely necessary to conduct this study from both male and female perspectives, which would be conducive to achieving gender equality in the whole DJ community. The average working year of experts was 4.33, and the longest had 8 years of experience. The interviews were conducted remotely via Tencent Meeting, and each interview lasted approximately 40 minutes. We carry out the one-to-one interviews from three aspects: (1) the current situation and implicit bias across the entire industry, (2) the use of VR technology in DJ performance, and (3) gender and avatar representation of DJs in social VR. Through in-depth analysis, we summarize a set of valuable information from expert responses.

First, all experts agreed that some injustices, stereotypes, and risk factors exist in the industry. For example, E2 and E3 pointed out that, '*music festivals are always male-dominated, female DJs have fewer opportunities and lower pay*'. E1 supplemented, '*female DJs are always deemed to acquire recognition on appearance rather than skills, even leading to intimidation and sexual harassment may happen*'. E2 further noted that some female DJs choose to cater to this social phenomenon, which may deepen misconceptions about this group.

Second, experts upheld a positive view of VR technology used to enhance the holistic experience in a live performance. For

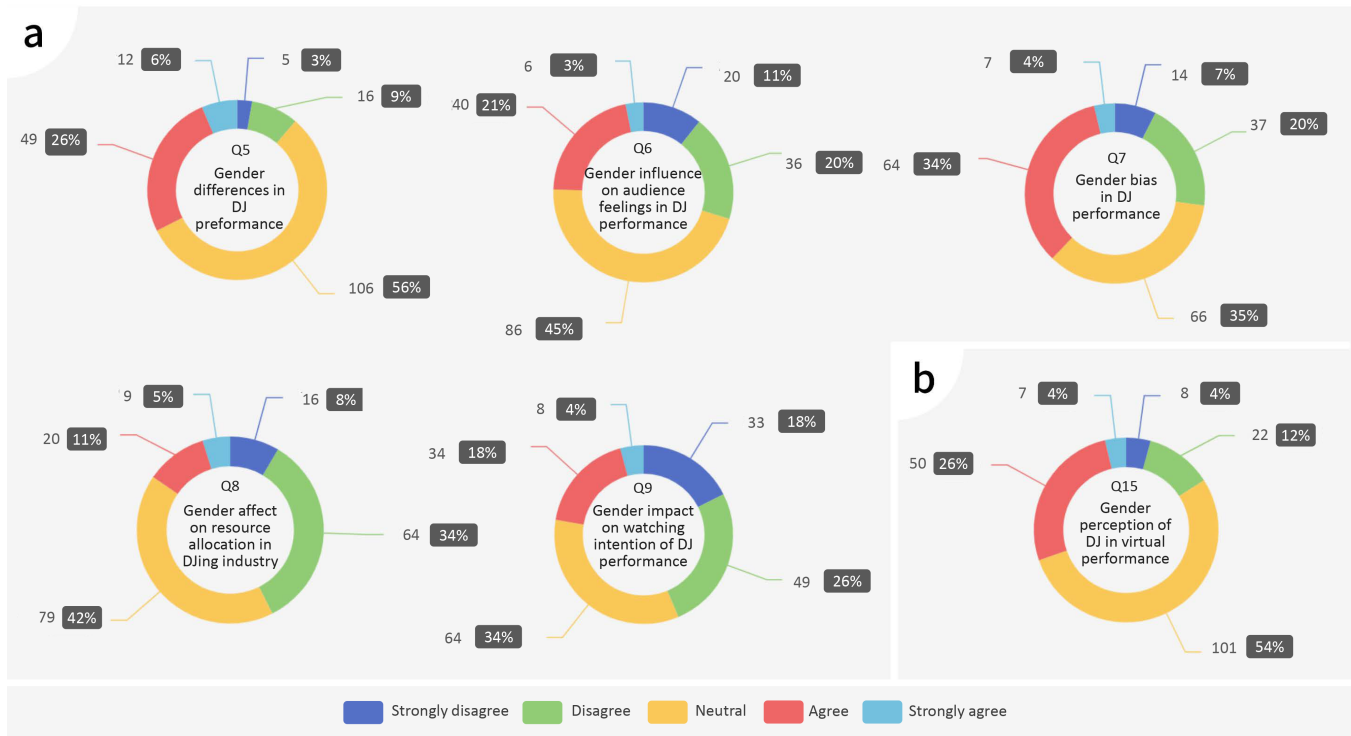


Figure 2: Survey results of gender factors in (a) DJs, and (b) virtual DJs.

example, E1 suggested that the space design and activity planning in VR environments could be more idealistic and attractive to align with the music theme beyond replicating onsite live performances. E2 supplemented, *'it can break out the geographical and temporal limitations to engage in this kind of event'*. On the other hand, some challenges have also been identified, particularly for DJ performers. E1 raised, *'it should have further discussed how to observe audience response and mobilize the atmosphere in VR'*. E2 pointed out the most significant issue in VR DJing, *'it is not easy to use VR controllers to simulate interaction with virtual devices in current VR DJ solutions, this lack of fluency and intuitiveness in manipulation, compare with the onsite experience'*.

Third, all experts acknowledged the visual representation of avatars has a potential impact on social interaction and gender perception in virtual performance. As E3 noted, *'since audiences have different expectations and responses for male and female DJs, this also would extend to their attitudes on virtual DJs'*. In response to that, E1 expressed that adding gender-neutral avatars could be a potential solution to avoid stereotypes. Additionally, E2 further suggested for avatar design, *'the visual representation of DJs should align with a specific musical context'*.

2.2 Online Survey

We target the general public for our online survey to obtain their attitudes and preferences in virtual DJs and social VR live house experiences. Participants took an average of 2 minutes to complete the survey. A total of 188 valid responses (124 females and 64 males) were received, aged between 17 - 33 ($M=21.13$, $SD=3.82$). Most

(72.87%) respondents hold a bachelor's degree or above. We used IBM SPSS Statistics for statistical analysis, as the result depicted in Figure 2.

According to the analysis, we see most respondents (71.67%) have been to the live house and a fifth (19.15%) of them watch the DJ performance more than once a month (Q4). The majority of responses reported that they perceived a significant difference ($M=3.25$, $SD=.82$) between male and female DJ performance (Q5), and they agreed that gender factors could influence subjective feelings in DJ performance ($M=2.87$, $SD=.97$) (Q6). Most of them ($M=3.07$, $SD=.99$) acknowledged that the performance of female DJs is often underestimated (Q7). The majority of responses ($M=2.96$, $SD=.94$) felt that there is no equality in resource allocation between males and females in the industry (Q8). However, most respondents ($M=2.56$, $SD=1.10$) didn't think that gender was a primary consideration for their choice of watch DJ performance (Q9). Additionally, we asked multiple-choice questions to elicit their specific requests about DJs and live house events. For the perceptions of DJ performance and competence, a majority of respondents emphasized the importance of interactivity for both male DJs (60.64%, Q10) and female DJs (70.21%, Q11), followed by skills for male DJs (57.45%, Q10), and appearance for female DJs (65.96%, Q11). We extracted five primary factors of DJ's skills and qualities from expert interviews to further investigate their importance. A Friedman test showed a statistically significant difference in ranking, $\chi^2(4) = 132.728$, $p < .001$. The factors that audiences are most concerned about in order are interaction, music style, appearance, remixing skill and the

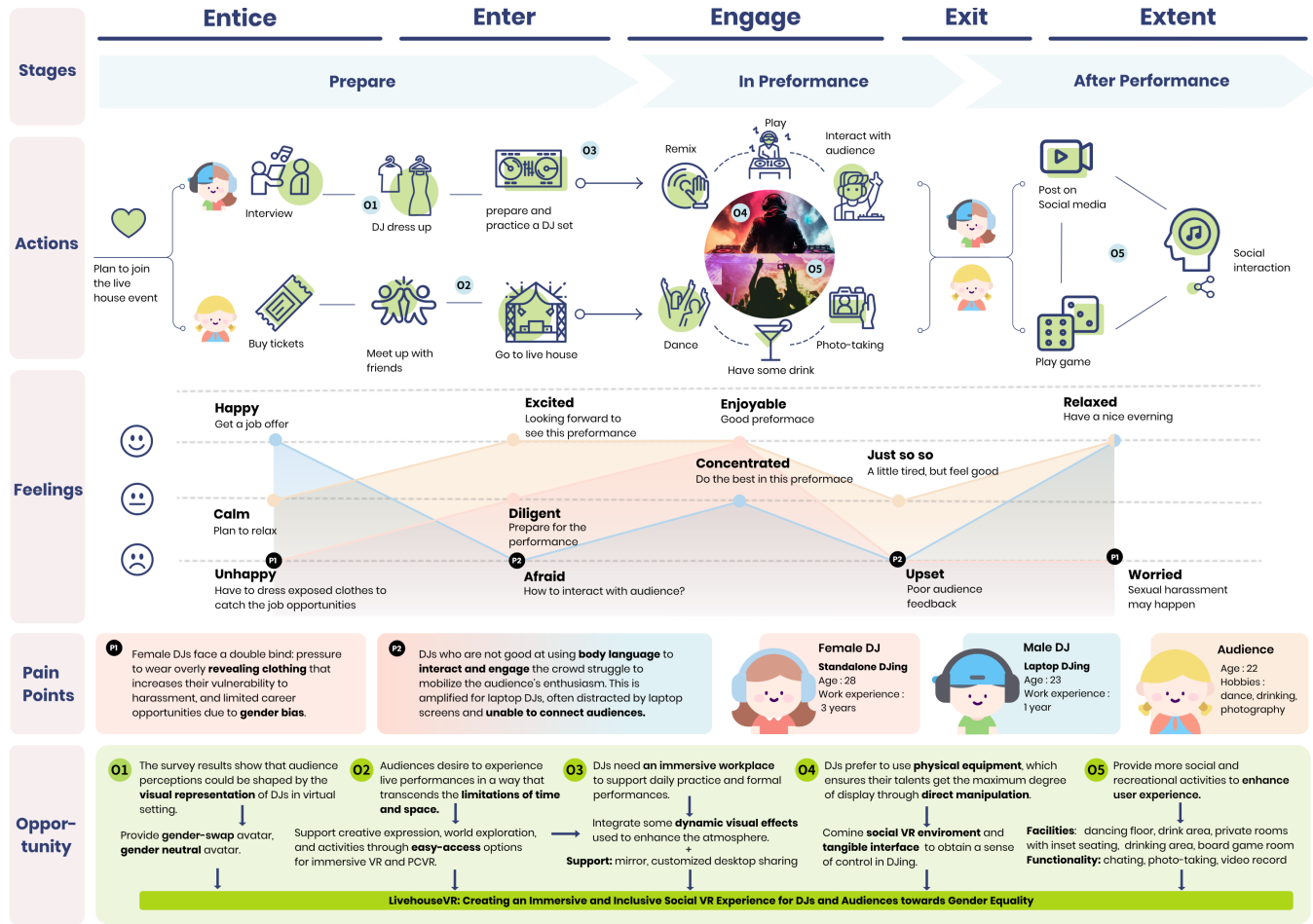


Figure 3: User journey map of a live house event in five stages (P = Pain point, O = Opportunity).

originality of DJ sets (Q12). Although nearly half (56.91%) of the respondents haven't heard about virtual DJ performance ($M=1.57$, $SD=.50$) (Q13), 49.47% of the responses had shown a moderately interested ($M=2.88$, $SD=.94$) in virtual DJs (Q14). Moreover, when asked about the gender perceptions of DJs in VR environments, more than half (53.72%) of the respondents kept a neutral attitude ($M=3.14$, $SD=.83$), still, a third (30.32%) of the respondents made clear that the visual representation of virtual DJs would affect their perceptions of the biological gender of DJs (Q15). For the live house event, most respondents would like to see some dynamic visual effects used to enhance the immersive atmosphere (70.21%), such as neon displays, cyberpunk aesthetics, laser shows, and particle effects, followed by diverse social interactions support (67.02%), such as broadcasting, private rooms, and recreational activities (62.77%), such as drinking, and casual games (Q16).

The survey results reveal several key findings: (1) there is a growing awareness of gender inequalities in the DJ industry and a desire for more inclusive practices; (2) audiences assess DJ performances based on a combination of factors, including interactivity, music selection, technical skills, and visual presentation, underscoring the multifaceted nature of this art form;

(3) virtual reality has the potential to reduce gender bias by concealing physical appearance, audience perceptions continue to be shaped by the visual representation of DJs in virtual setting; (4) there is considerable interest in virtual DJing experiences, suggesting a viable market for immersive and interactive VR platforms that can replicate and enhance the social dynamics of real-world performances; (5) audience expectations for virtual live house events are notably high, with a strong demand for dynamic visual effects, diverse social interaction features, and engaging entertainment activities that foster a sense of community and shared experience.

3 IDENTIFY THE DESIGN GOALS IN A USER JOURNEY

To further define the design goals, we summarize the preliminary research findings from expert interviews and only survey into a user journey map to present an experiential user scenario. We followed the Experience Design Model [16] to define the five stages of the user journey. Fig.3 shows the whole process for a live house event that both DJs and audiences' experience in a social environment.

DG1: Facilitate engaging cross-platform live house experiences. To address the demand for immersive music experiences that transcend physical limitations, we intend to use traditional PC and emerging VR as technical means of providing inclusive access (O1). In response to this, several factors need to be considered, such as freely available, user privacy, cross-platform compatibility, and extensive support of user-generated content [17].

DG2: Support self-expression for DJs through the visual representation of avatars. Building on DG1 and addressing issues raised in preliminary research about gender bias (P1), we hope the avatar identification and identity confirmation extend to the real world [9] to deal with the gender sensitivity of DJs. This would help to facilitate self-expression, contributing to a more equal DJ community (O2).

DG3: Create an immersive atmosphere adapted to the music theme. Environments play a notable role in shaping music creation and experience, which brings the human sense of place and music expression [2]. By combining visual effects aligned with the musical content, we aim to create a holistic experience that goes beyond the fidelity and dynamics of live DJ performances (O3).

DG4: Elevate DJ skills through embodied interactions and effective feedback. To support daily practice and formal performances in workplace (O3), we would map and optimize the DJ actions between audiences, visual screens, and devices. This includes reflecting the positions of physical DJ equipment in sync (O4), customized virtual setups, and real-time feedback of body language and reactions (P2).

DG5: Provide multi-purpose spaces with a variety of facilities to strengthen social bonds. Participants are often motivated by different purposes and needs to engage in social activities [6]. In this case, we would offer both public and private spaces with a rich array of recreational facilities to foster communication, shared experiences, and community building among the public (O5).

4 LIVEHOUSEVR

We derived main system features based on the discussion of design goals, and ultimately formed the design, a public world named 'LivehouseVR' in VRchat platform. Here, we illustrated how to implement LivehouseVR in a five-stage user journey and conducted evaluations with a DJ practitioner and eleven volunteers.

4.1 Design and Implementation

Entice: VRchat world. In the *entice* stage, LivehouseVR is built in Unity with VRChat Creator Companion [DG1]. We use Udon scripting language supported by VRChat SDK to design interaction techniques and methods. As a result, DJs and audiences could access the world using the VR-HMD or the PC version of VRChat. With its built-in stage (see Fig.4a), sound and lighting system, and LED walls, this venue can make performances more visually spectacular [DG3].

Enter: waiting room. In the *enter* stage, we create a virtual waiting room for DJs (see Fig.4b). They can select roles from avatar

pedestals placed in this space to promote a strong sense of self-identity for combating gender inequality [DG2]. Besides, we offer practice settings for DJs to refine their skills before they complete the identity registration and take the stage. In addition to essential manipulation support, we incorporate mirrors to facilitate real-time feedback on gestures and dynamic motions in practice, enabling DJs to convey enthusiasm and engage audiences effectively [DG4].

Engage: DJ console. In the *engage* stage, physical DJ equipment is used as a tangible medium to realize direct control in virtual performance (see Fig.4c). To facilitate MIDI mapping and synchronization, we distribute MIDI signals via two virtual ports: one designated for Rekordbox, a music visualizer, and the other for the virtual equipment in VRChat. This setup enables the simultaneous control of both Rekordbox and the virtual DJ setup in VRChat by sending MIDI data [DG4]. In this process, the MIDI output from the controller has been translated to align with the requirement of Rekordbox. Furthermore, we establish ownership protocols for in-game objects within VRChat. Only the object owner (DJ) can set synchronized variables associated with it, which can then be linked to the DJ console (see Fig.4d). DJs can move and resize relevant virtual objects, including the console, screen, and equipment. Their performances are mirrored on a top screen, allowing them to observe body movements easily (see Fig.4e).

Exit: social facilities. In the *exit* stage, we provide a variety of areas to cater to various social preferences. Audiences can engage in embodied interaction to relax on the dancing floor with synchronized lighting. The virtual bar area fosters a relaxed atmosphere conducive to conversation (see Fig.4f). For those desiring a more intimate experience, private rooms are available, offering a secluded setting. Additionally, some mini-games, such as rolling dice, photo-taking (see Fig.4g), facilitate playful interactions and encourage self-expression in social contexts. These varied spaces and activities enhance the user experience and address a broad spectrum of social needs [DG5].

Extend: live streaming. In the *extend* stage, DJ performance is transmitted across the VRchat world to be shown on virtual screens, ensuring that people in different areas can view the performance. Furthermore, we take advantage of live streaming to facilitate instantaneous interaction for broader audiences (see Fig.4h). A bidirectional communication channel have been created with a popular video platform, Bilibili. External audiences can react, comment, and engage in real time [DG1]. All these messages are sent back and projected on a virtual screen for DJs to obtain immediate feedback from the broader audience, forging deeper connections with audiences worldwide.

4.2 Evaluation

In this study, twelve participants aged between 15 to 28 ($M=22.25$, $SD=3.39$; 5 males and 7 females) were recruited using snowball sampling, including a female DJ and eleven audiences interested in a live house event. We first set up a WeChat group for our participants to introduce our experiment. Then, participants were invited into the VRChat world and guided to select avatars and adjust the other settings. The event lasted about two hours, and two researchers guided participants through the whole process. DJ

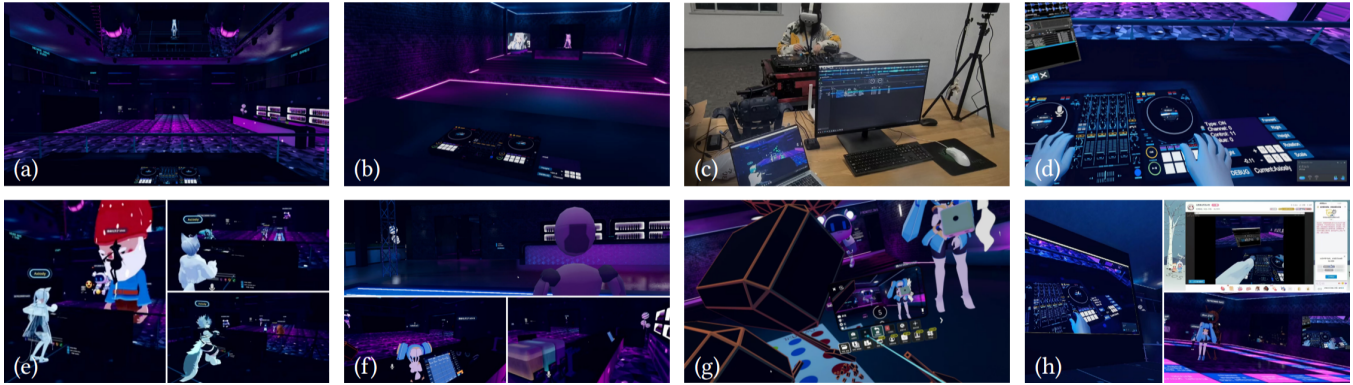


Figure 4: An overview of LivehouseVR in VRChat.

was led to the preparation room and then registered as DJ, while audiences directly went to the indoor stage for social interactions and to enjoy the DJ performances. After the performance, participants were free to move around and participate in various activities in each area. At the end of the event, all participants were asked to fill in a questionnaire and conduct a semi-structured interview for approximately 30 minutes.

The System Usability Scale (SUS) scores provided by participants are presented in Figure 5. We adopted the validated criteria from Bangor et al.[1] to interpret these scores. The average SUS score returns 78.96 with an SD of 15.24, indicating that the system falls within the ‘good’ to ‘excellent’ range. Furthermore, three participants rated LivehouseVR as ‘best imaginable’, suggesting high levels of usability and user satisfaction. We then categorized the semi-structured interviews into five dimensions, including general questions (e.g. is there anything else you’d like to share about your experience? Do you have other suggestions?) and some specific questions asking DJ (e.g. did you feel the controls were intuitive and responsive?). Participants were specifically asked for suggestions on how to improve LivehouseVR.

General Impressions of LivehouseVR. Participants expressed positive feedback regarding the overall design of LivehouseVR. For instance, P1 admired the accessibility of the virtual platform, highlighting its ability to provide a ‘fast and smooth’ experience. Notably, DJ performance is one of the most popular features (N=7), since its creativity and interactivity in terms of avatar customization and social communication. Apart from that, participants (N=5) also confirmed the social benefits and inclusion that LivehouseVR brought. P9 noted that ‘LivehouseVR allows anyone to participate in virtual performances regardless of time and location, providing a more flexible environment to relax’. In particular, many participants (N=7) mentioned that room settings and social activities are very realistic, and P6 further suggested that ‘providing more informative guidance and navigation could help users take full use of these facilities’.

Comparison with Onsite Experience. Concerning the question of ‘compared to attending an onsite concert or club, what did you enjoy most about LivehouseVR?’, participants evaluate LivehouseVR in three aspects. (1) Flexibility (N=4). For example, P2 noted, ‘LivehouseVR avoids the crowded environment typically found in offline settings, and it offers unrestricted freedom of movement,

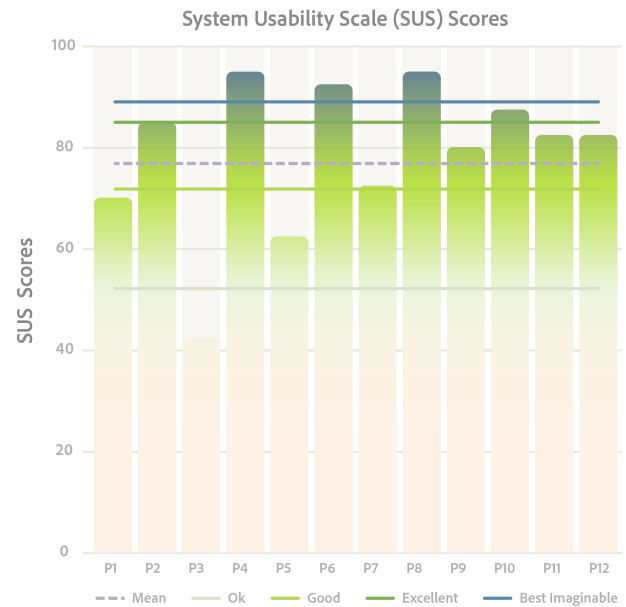


Figure 5: The SUS scores of LivehouseVR.

allowing users to enjoy the event from any viewpoint’. (2) Security and privacy (N=3). As P8 emphasized, ‘in the virtual space, no one knows who you are, which reduces self-perceived concerns regarding appearance and inspires me to bravely express myself’. (3) Beyond reality (N=4). LivehouseVR breaks out the constraints of onsite venues and opens more possibilities in a virtual world supported by dreamlike scenes and creative interactions. P10 further pointed out, ‘compared to traditional online performance, LivehouseVR facilitates embodied interactions that enhance user engagement and foster a more active performance, in contrast to passive viewing’.

Gender Bias and Perception in DJ Performance. All participants agreed that LivehouseVR has the potential to mitigate gender bias, which also enhances audience and DJ interaction. For example, P9 noted, ‘the interaction between the audience and the DJ is more active in virtual environments, free from the influence of

physical appearance and other real-world factors. This view was supported by P3 and P4, who stated, *'by hiding the DJ's physical appearance through avatars, LivehouseVR encourages the audience to evaluate the DJ based on their skills and the performance they created'*. This focus shifts from the DJ's identity to the quality of the performance, fostering a more inclusive and equal environment. From the DJ perspective, LivehouseVR offers a range of advantages that enhance their performance and experience. P6 emphasized that *'LivehouseVR provides a more creative and immersive environment, as it is specifically designed to align with music themes, enabling DJs to concentrate on their performance'*. Moreover, P6 mentioned that *'the mirror provided in LivehouseVR stage can help DJs observe themselves, adjust the facial expressions and body language in real-time, and behave in a way that is more in keeping with the avatar'*. P2 supplemented, *'this promotes self-expression and encourages DJs to explore new ways of interacting with audiences'*.

5 DISCUSSION

We worked towards exploring the potential of social VR in live music performances, which are guided by our design goals extracted from a five-stage user journey and followed by the Experience Design Model [16]. Our design incorporates a rich array of features to mitigate gender bias and foster inclusive interactions. The evaluation showed that LivehouseVR provides a more immersive way for DJs and audiences to engage in performance, social interactions, and recreation. In summary, our findings suggest several key takeaways regarding the design of social VR experiences.

The primary goal of this study is to investigate how a curated social VR environment could reduce the impact of gender bias in live DJ performances. Our findings indicate that LivehouseVR is successful in this regard. Participants reported that the virtual performance, where DJs are represented by customizable avatars, shifted the focus from physical appearance to musical skill. Our finding is in line with previous work showing that avatars can influence the audience's perception of DJ by reducing the impact of gender bias, thereby shifting focus toward DJ's technical skills [4]. Thus, providing diverse avatar options, including non-human avatars, is an effective solution to mitigate gender bias. This feature promotes free self-expression without the constraints of traditional gender presentation in the real world.

Another key finding indicated that the virtual performances in LivehouseVR complement traditional onsite music events, not replacements. In fact, virtual performances in the form of social VR experiences can never replace the meaningful social experience in onsite venues [19]. LivehouseVR should adapt to users' mental models of physical attendance. This means respecting the existing context with daily settings, such as providing communal social spaces and exclusive alcohol. Social VR interfaces such as LivehouseVR can help attract and motivate users to attend a physical live house for the first time and revisit them to experience the music performance in a traditional setting. This holds great value as a gateway to the world of live music performance.

However, we acknowledge that some limitations and challenges still exist in this study. First, the number of DJs and audiences was

limited in user evaluation, and each participant came from a relatively homogeneous background. Furthermore, we could not compare experiencing LivehouseVR to a control group using another pre-existing solution as a valid baseline to compare with the prototype since it offers an entirely new technology-supported social experience. Additionally, while this virtual space design indicates the potential for promoting gender equality through social VR, it cannot definitively solve the problem. Instead, this work can be concluded as an exploratory study that falls into the category of discursive design [18], which seeks to generate discussions on such a social and psychological topic related to gender bias in DJ community for future development.

6 CONCLUSION AND FUTURE WORK

In this paper, we propose LivehouseVR, a social VR solution to create inclusive and engaging spaces for live music performances, contributing to gender equality. We present our research and requirement gathering through expert interviews and online surveys, followed by our system prototype. We evaluated our design implementation from both DJ and audience perspectives. This research provides insights into relevant work exploring the intersection of technology, gender, and creative expression. As social VR platforms continue to evolve, future research should focus on refining the technical aspects of social VR performance environments, exploring new forms of interaction, and investigating the long-term impact of these platforms on the music industry and social interaction more broadly.

Acknowledgments

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